Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

- I will use dataclasses to make instances of vehicles
- I will use arithmetic operators such as +, -, * and /
- I will use several if statements
- I will loop over the list of locations the user can travel to
- I want to improve on using drafter while completing this project

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect designing my data classes and implementing helper functions to be easy for me because I'm familiar with these concepts and have practiced them in homework assignments. I expect building my root functions to be hard because they all have to use my helper functions and the dataclasses I design.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think the thing that will take the most time and effort is displaying and updating the roadmap page, which shows where the user's car is currently located. As this is going to be a difficult part of the project, I want to start with an easier part of the project, such as designing my data classes.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

While designing, it may be distracting to think about how far I am from achieving my goal of completing the project. I will minimize this distraction by telling myself to trust the process, knowing that I'm making progress.

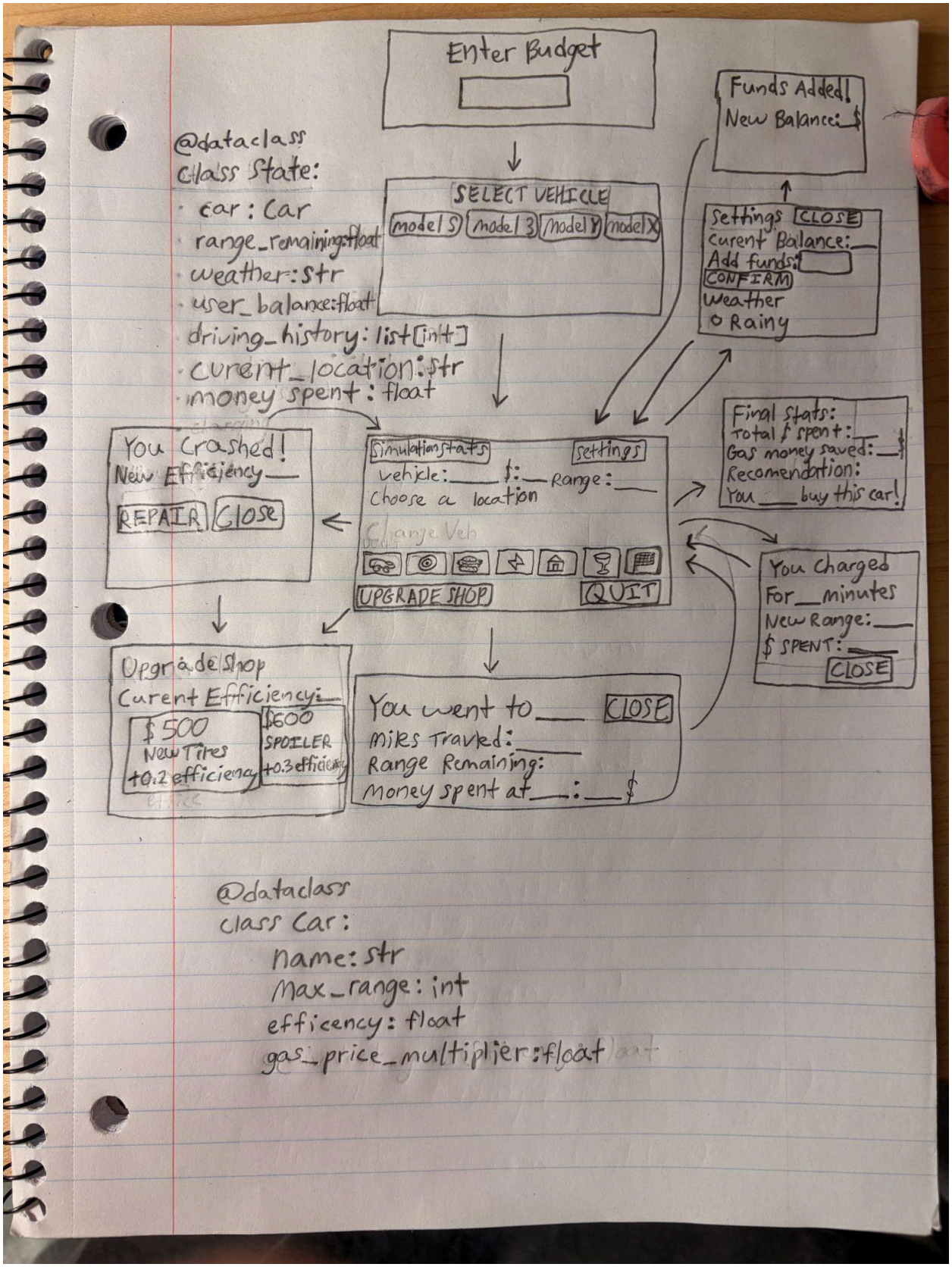
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.



2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**Allows users to simulate driving an electric Tesla vehicle. Based on user input, it provides the user with real-life stats about driving the vehicle.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	● car: Car ○ The users car (attributes listed in “third dataclass”) ● money_spent: float ○ Will increase as the user goes to locations and charges their vehicle ● user_balance: float ○ The money the user has left in their account. Decreases as they spend money, increases when they add. ● locations: list[Location]
Second Dataclass: Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	<u>The Location Dataclass:</u> ● name: str ○ The name of the location. Never will be modified. ● price_to_attend: float ○ The price to attend the location. Never will be modified. ● mile_marker: int ○ How far along the 2D list of locations in which the location is located. Never will be modified.
Extra Third Dataclass:	<u>The Car Dataclass</u> ● name: str ○ The name of the car. Never will be

	modified. ● max_range: int ○ The maximum range of the car on a full charge. Never will be modified. ● efficiency: float ○ The current efficiency of the car. Increases as the user purchases upgrades, decreases when the user crashes. ● gas_price_multiplier: float ○ A constant multiplied to the money spent on charging, to find the equivalent cost of gas ● miles_driven: int ○ The miles the car has driven. Increases as more locations are traveled to. ● range_remaining: int ○ The current range left in the vehicle's battery. Decreases as more locations are traveled to.
Routes (minimum 7) The names of your route functions	● index ● display_vehicle_options ● update_map ● display_settings ● show_added_funds ● show_final_stats ● show_destination_recap ● show_charging_recap ● show_car_crashed
Pages (minimum 5) A brief description of the pages in your site	● Initial budget page ○ Where are the user enters their initial budget ● Vehicle selecting page ○ Where are the user selects their vehicle ● Roadmap page ○ displays the interactive map of where the user can travel to as well as other options ● Settings page ○ Where are the user can add money to their account

	• Upgrade Page ○ Where are the user can purchase upgrades for their vehicle • Charging completed page ○ Displays the stats of a charging session • Destination attended page ○ Displays the results of attending a destination • Crashed page ○ Informed the user of their vehicle crashed • Final stats page ○ displays the final stats of the user's simulation
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	• Enter starting budget box - TextBox ○ Where are the user enters their initial budget • Car choice dropdown menu - SelectBox ○ Allows the user to pick which car they want to test • Add funds amount box - TextBox ○ Where are the user types the amount of funds they want to add to their account
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	• An if statement will be implemented to only include car options in the car choice dropdown menu that the user can afford based on their starting budget • An if statement will be implemented to display a warning sign on locations within the location map that a user either does not have enough range to get to or does not have enough money to attend • An if statement will be implemented and used each time a user goes to a location to evaluate if a randomly generated number is within a certain range. If it is, the user's car will crash • An if statement will be implemented to check if the user selected a location to travel to that they don't have enough range to get to. If the statement evaluates to false, the user's car

	will be towed
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	When the update_map route function is called, a for loop will be implemented to loop over the list of locations and mark each one that a user does not have enough range to get to or enough money to attend with a warning sign.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.